A Shift-Share Measure of Regulatory Scope

TEACHING NOTE

ACCTG 597A · Week 10

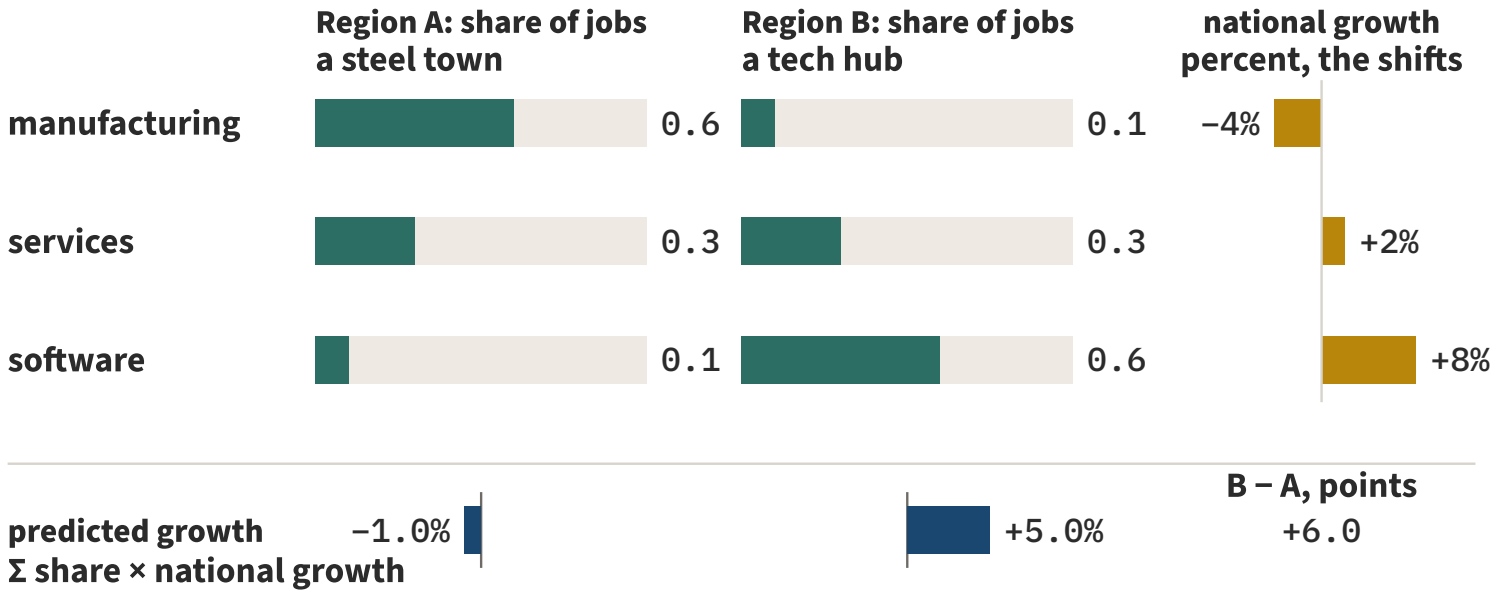
On Breuer, Leuz, and Vanhaverbeke (2025; BLV), "Reporting Regulation and Corporate Innovation," *Journal of Accounting and Economics* 80(1): 101769. The paper's numbers carry their page; the simulation is this note's.

A shift-share instrument predicts a local change from fixed shares and common shocks

- Bartik (1991) asks what local labour demand does to local employment and wages. Actual local employment growth also reflects local supply, so the instrument is the growth a region would have had if each of its industries had grown at the national rate: initial industry shares times national industry growth.

> The shares are exposure, fixed before the shocks arrive; the shifts are common to every region; one region's predicted growth differs from another's only through its shares. Blanchard and Katz (1992) use the same construction for state employment.

Numbers on the stage are illustrative. Glosses of Bartik (1991) and Blanchard and Katz (1992) follow the verified literature report for this note.



national software growth **+8%** · predicted growth: A **−1.0%**, B **+5.0%** · the gap moves by (0.6 − 0.1) × the shock

national software growth, percent +8

Stylised: two regions with illustrative industry shares and three national growth rates. Bartik (1991) and Blanchard and Katz (1992) use exactly these ingredients, a region's initial industry employment shares (teal, the shares) and national industry employment growth (gold, the shifts), to predict the region's employment growth (navy, the instrument).

The instrument is exposure times shock, so its variation is a comparison across exposure

The instrument for unit l, with shares s_{lk} fixed in a base period and shocks g_k common to all units

$$z_l = \sum_k s_{lk}\, g_k, \qquad x_l = \pi\, z_l + v_l, \qquad y_l = \beta\, x_l + \varepsilon_l$$

> Relevance: the actual change tracks the predicted change ($\pi \neq 0$). Exclusion: z_l moves y_l only through x_l. With unit and time effects, what identifies β is the interaction of exposure with the shocks: units with more exposure to a shock change more, a difference-in-differences across exposure.

Instance	Unit	Shares (exposure)	Shifts (the shocks)
Bartik (1991); Blanchard and Katz (1992)	region or state	initial industry employment shares	national industry employment growth
Card (2001)	city	past settlement of each origin group	national inflows by origin
Autor, Dorn, and Hanson (2013)	commuting zone	initial industry employment shares	Chinese import growth in other high-income countries
Breuer, Leuz, and Vanhaverbeke (2025)	country-industry	the industry's Europe-wide size distribution	a country's size thresholds

Glosses from the verified literature report for this note; the last row is this note's reading of the paper (slides 09 and 10).

Identification rests on the shares or on the shifts

Shares as exposure (Goldsmith-Pinkham, Sorkin, and Swift 2020)

The shares s_{lk} are the instruments; the shifts only weight them. The condition: the shares are uncorrelated with the outcome's residual given the effects, so a unit's exposure does not predict how its outcome would have changed without the shock. Diagnostics: Rotemberg weights (which shares carry the estimate); balance of high-weight shares on observables and on pre-period trends; dispersion of the just-identified estimates.

Shifts as shocks (Borusyak, Hull, and Jaravel 2022)

The shifts g_k are as good as random across many shocks; the shares may be endogenous. The condition: a law of large numbers over shocks, with an effective number of shocks (an inverse Herfindahl of shock weights) that is large. Diagnostics: a shock-level regression; shock-level balance; clustering at the level of the shock. Borusyak and Hull (2023): a formula-based exposure, simulated eligibility included, can be recentered on its expectation over counterfactual shock draws.

A paper has to say which route it takes, because the two ask different questions of the data: of the shares, whether exposure predicts counterfactual trends; of the shifts, whether there are enough of them and whether they were assigned without regard to the units. The second half of this note asks both of Breuer, Leuz, and Vanhaverbeke.

The variant in Breuer, Leuz, and Vanhaverbeke (2025)



Forcing private firms to publish statements could raise or lower innovation

In the EU every limited-liability firm must publish full financial statements unless its country exempts it as small. Breuer, Leuz, and Vanhaverbeke (2025; BLV) ask what the mandate does to innovation, measured in the Community Innovation Survey as a new or significantly improved product, process, organisational method, or marketing method (p. 7).

The room's bet: the mandate could raise the share of innovating firms, through what rivals and suppliers learn from the reports, or lower it, through the proprietary costs of disclosing.

Sample: up to 26 countries, 2000 to 2018, about 7,100 country-industry-years (pp. 3, 11).

- **BLV find a decline: a 10 point rise in the mandate's scope is associated with 1.8 points fewer innovating firms, about 5 percent of the mean, and the coefficient on total industry spending is not distinguishable from zero (Table 2, p. 12). Both predictions had a theory; the answer rests on the design.**

−0.182

on the share of innovating firms (t = −2.96); Table 2, column 1

−1.706

on average log innovation spending (t = −2.03), about 16 percent lower per 10 points; Table 2, column 3

0.233

on log total industry spending (t = 0.25); Table 2, column 4

The mandate reaches a firm only above its country's size thresholds

- **A country exempts firms below thresholds on total assets, sales, and employees, within EU maxima; the thresholds differ across countries, and BLV call them persistent over time (p. 6; fn. 15, p. 8; Table OA1).**

Criterion	EU maximum, most of the sample (fn. 8, p. 6)
total assets	about 4 million euros
sales	about 8 million euros
employees	50

Exempt firms typically file an abridged balance sheet and withhold the income statement and the management report (p. 2). This note collapses the three criteria to one firm size.

> *"A central feature of the threshold-based regulation in Europe is that a given country's exemption thresholds affect industries in different and, importantly, predictable ways. For example, a regulation that exempts firms below 50 employees from full reporting affects labor-intensive industries more strongly than capital-intensive industries."*
>
> Breuer, Leuz, and Vanhaverbeke (2025, p. 8)

The two industries of this note are BLV's pair: a labour-intensive industry with many small firms and a capital-intensive industry with large firms, facing the same employee threshold.

The actual share of mandated firms responds to innovation itself

- **If the treatment were the share of firms actually above the thresholds, it would move with innovation: technology shocks grow firms into the mandate, and industrial policy shapes firm sizes in particular country-industries (p. 2). In the simulation, OLS on that share reports the wrong sign.**

2.77

OLS of innovation on the actual share when an innovation shock also grows the cell's firms (λ = 0.5); the paper's fixed effects; mean over 300 draws

−0.50

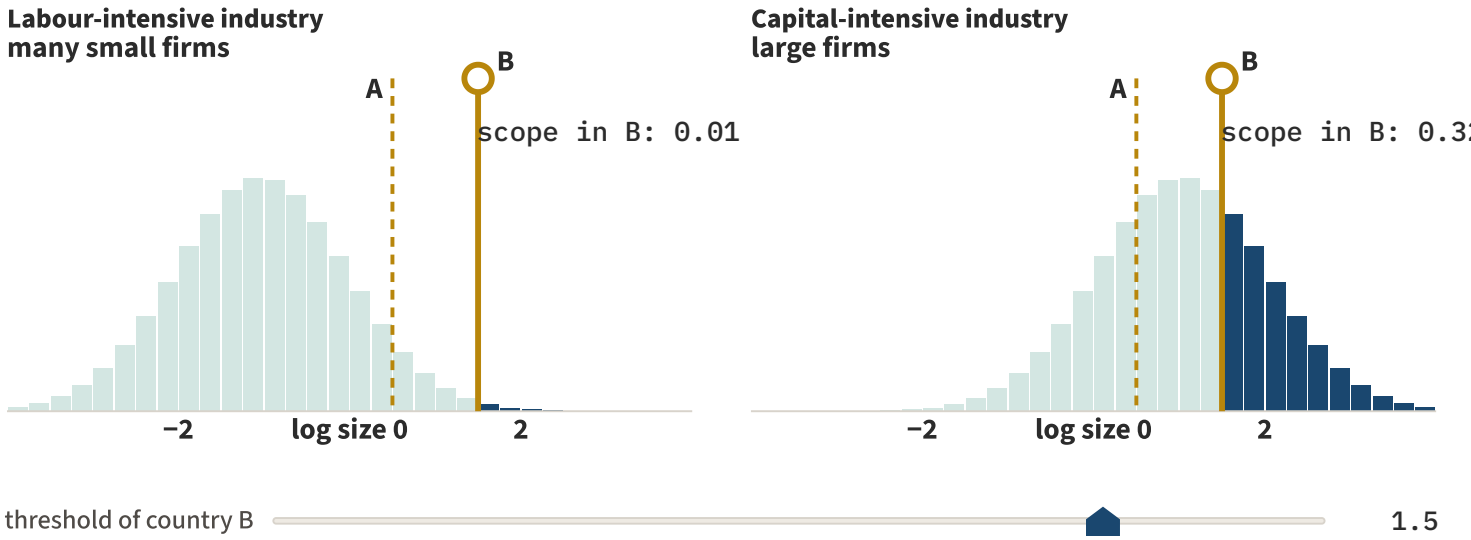
the causal effect of the mandate's scope, by construction; OLS equals it when sizes do not respond (λ = 0: −0.49)

> *"This approach alleviates concerns about endogenous firm-size differences, be it because of reverse causality (e.g., technology shocks causing firms in certain industries to grow above the thresholds), or omitted factors correlated with firm sizes in certain countries (e.g., industrial policies)."*
>
> Breuer, Leuz, and Vanhaverbeke (2025, p. 2)

Simulation (seed 2025; appendix: the data-generating process): the first channel is built in with strength λ; the second is the kind of factor the measure leaves out by construction and is not simulated.

A fixed size distribution turns a threshold into a simulated scope

- Applying a country’s threshold to a fixed industry size distribution gives the share of that distribution above it; the scope responds to the threshold in proportion to the industry’s density there, so the same threshold is a different treatment in each industry.



simulated scope	country A	country B	B − A
small-firm industry	0.06	0.01	−0.06
large-firm industry	0.70	0.32	−0.38
small − large	**−0.64**	**−0.31**	**+0.33**

Stylised: the simulation's two example industries, log size normal with means −1.03 and 1.03 and standard deviation 1 (teal), and a single scalar threshold in place of BLV's three criteria. The simulated scope is the share of each distribution above a country's threshold (navy); country A's threshold (gold, dashed) is fixed at 0.5. BLV's own intensities average 0.279 with an interquartile range from 0.089 to 0.319 (Table 1, p. 11).

BLV regress innovation on the simulated scope, a shift-share object

BLV's estimating equation (p. 8): country-year and industry-year effects, two-way clustering

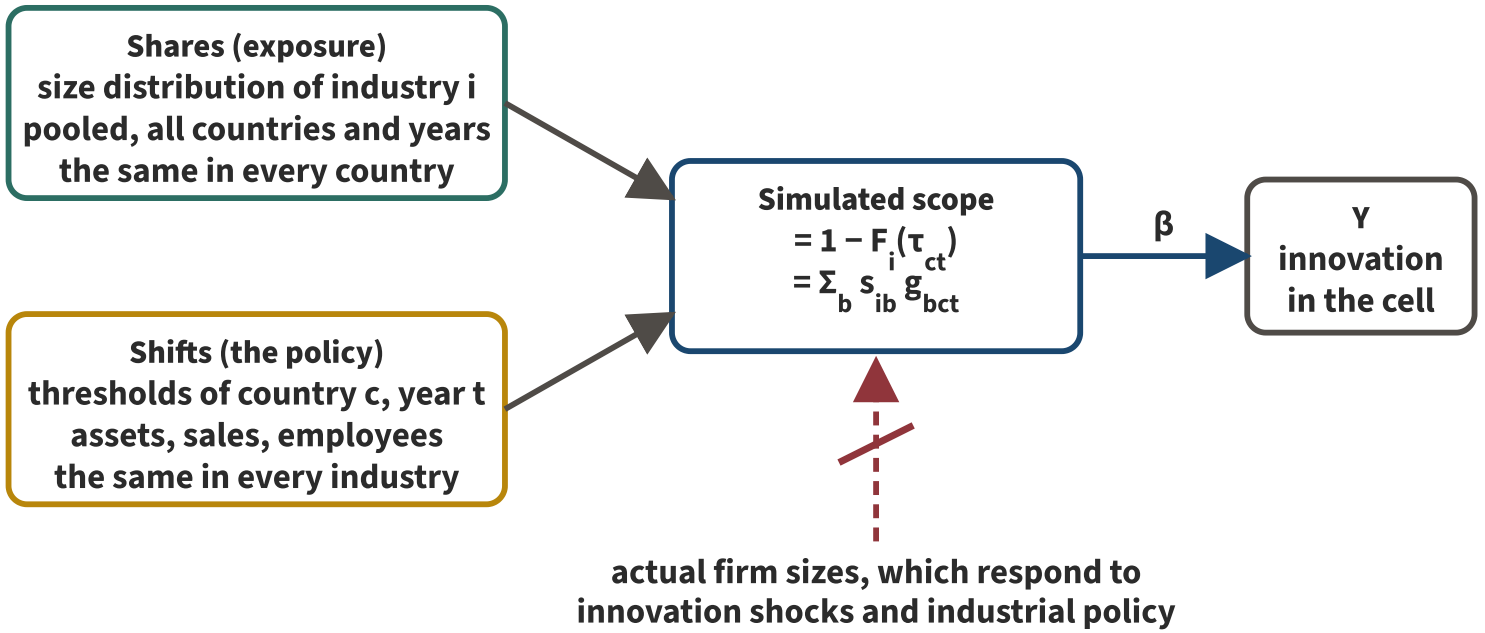
$$Y_{c,i,t} = \beta\,\text{Reporting}_{c,i,t-1} + \alpha_{c,t} + \delta_{i,t} + \varepsilon_{c,i,t}$$

The regressor (p. 9), written by this note as a sum over size bins

$$\text{Reporting}_{c,i,t-1} = 1 - F_i(\tau_{c,t-1}) = \sum_b s_{ib}\, g_{b,c,t-1}$$

> One object, several names: BLV's Reporting, regulatory intensity, standardized share, and simulated instrument; this note's simulated scope. s_{ib} are the **shares**, g the **shifts**. Against the canonical instance of slide 02, the roles are reversed: industries carry the shares and countries the shocks.

BLV call the measure "a variant of the Bartik instrument (Breuer, 2022; Goldsmith-Pinkham et al., 2020)" (p. 2) and a simulated instrument after Currie and Gruber (1996) (p. 9). Appendix: the algebra.



The measure takes an industry's Europe-wide size distribution (teal) and a country's thresholds (gold) and returns the share of that fixed distribution above the thresholds (navy), which BLV regress innovation on; the firms' own size responses (maroon, struck through) never enter it. Stylised to one scalar threshold; BLV apply three criteria. Drawn for this note from BLV pp. 2, 8, and 9.

Inside the fixed effects, the threshold levels identify the effect

- **Country-year effects remove what is common to a country's industries in a year, and industry-year effects remove what is common to an industry across countries; what remains of the measure is the country-by-industry interaction, and BLV state that it comes from threshold levels, not changes.**

> *"This design does not rely on changes in countries' thresholds over time, but instead compares differences in innovation for industries with different size distributions in countries with different exemption thresholds."*
>
> Breuer, Leuz, and Vanhaverbeke (2025, p. 2)

In the simulation three of ten countries change their threshold once. This note also treats the simulated scope as an instrument for the actual share, which BLV do not do: the first stage is the regression of the actual share on the simulated scope inside the effects. Appendix: the algebra.

83%

of the variance of the demeaned measure is explained by its threshold-level part under the paper's effects (one draw)

F ≈ 460

first stage of the actual share on the simulated scope under the paper's effects (mean over 300 draws)

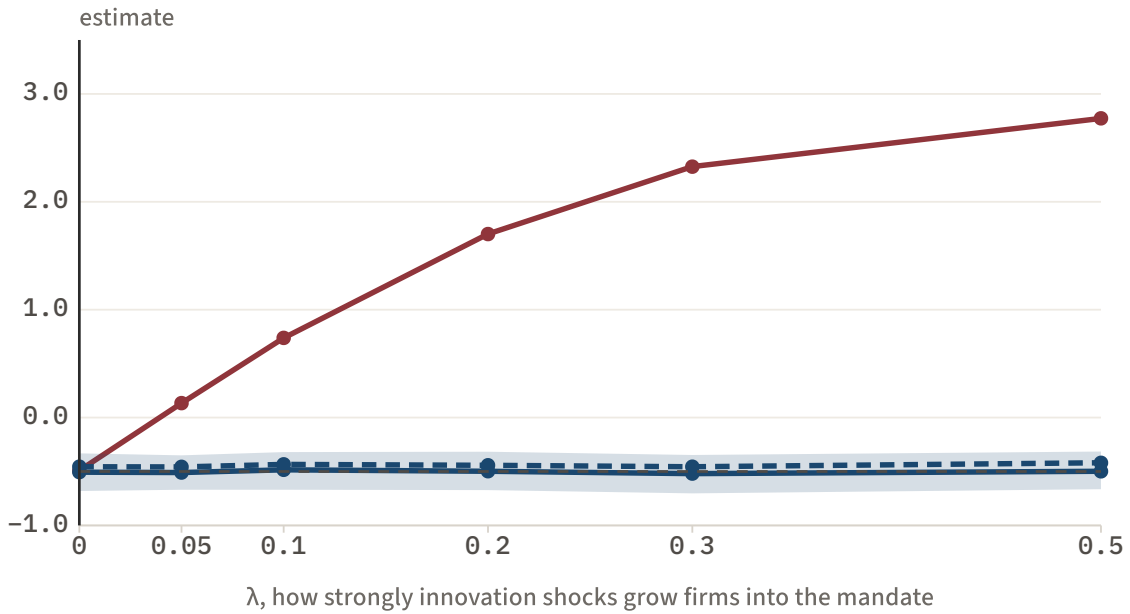
F ≈ 45

the same first stage after adding country-industry effects, which keep only the three threshold changes; the estimate's spread across draws is then four times wider (slide 15)

The simulated scope removes the reverse-causality bias

- **As innovation shocks grow firms into the mandate more strongly, OLS on the actual share drifts from the causal effect to 2.77, while BLV's reduced form on the simulated scope stays near −0.42 and 2SLS near −0.50.**

In the simulation the simulated scope omits the firms' responses, so its covariance with the innovation shock is zero by construction; BLV's pooled distribution contains the sample years' sizes (appendix: pooled shares). The reduced form sits above the causal effect because the first stage is below one, 0.90 at λ = 0 and 0.85 at λ = 0.5, which slide 14 takes up; its F falls from about 1,300 to about 470.



λ = 0.00 · OLS **−0.49** · reduced form **−0.46** · 2SLS **−0.51** · first-stage F ≈ 1,315

λ 0

OLS on the actual share · reduced form on the simulated scope (BLV's estimator) · 2SLS · 2SLS interquartile range · causal effect −0.50

Simulated (seed 2025, 300 draws per point): 30 industries, 10 countries, 8 years, 50 firms per cell; the causal effect of the mandate's scope is −0.50 by construction; country-year and industry-year effects, as in the paper. Lines are means across draws; the reduced form is BLV's estimator, 2SLS treats the simulated scope as an instrument for the actual share.

A reduced form, and an exogeneity claim



The coefficient is a reduced form per unit of statutory scope

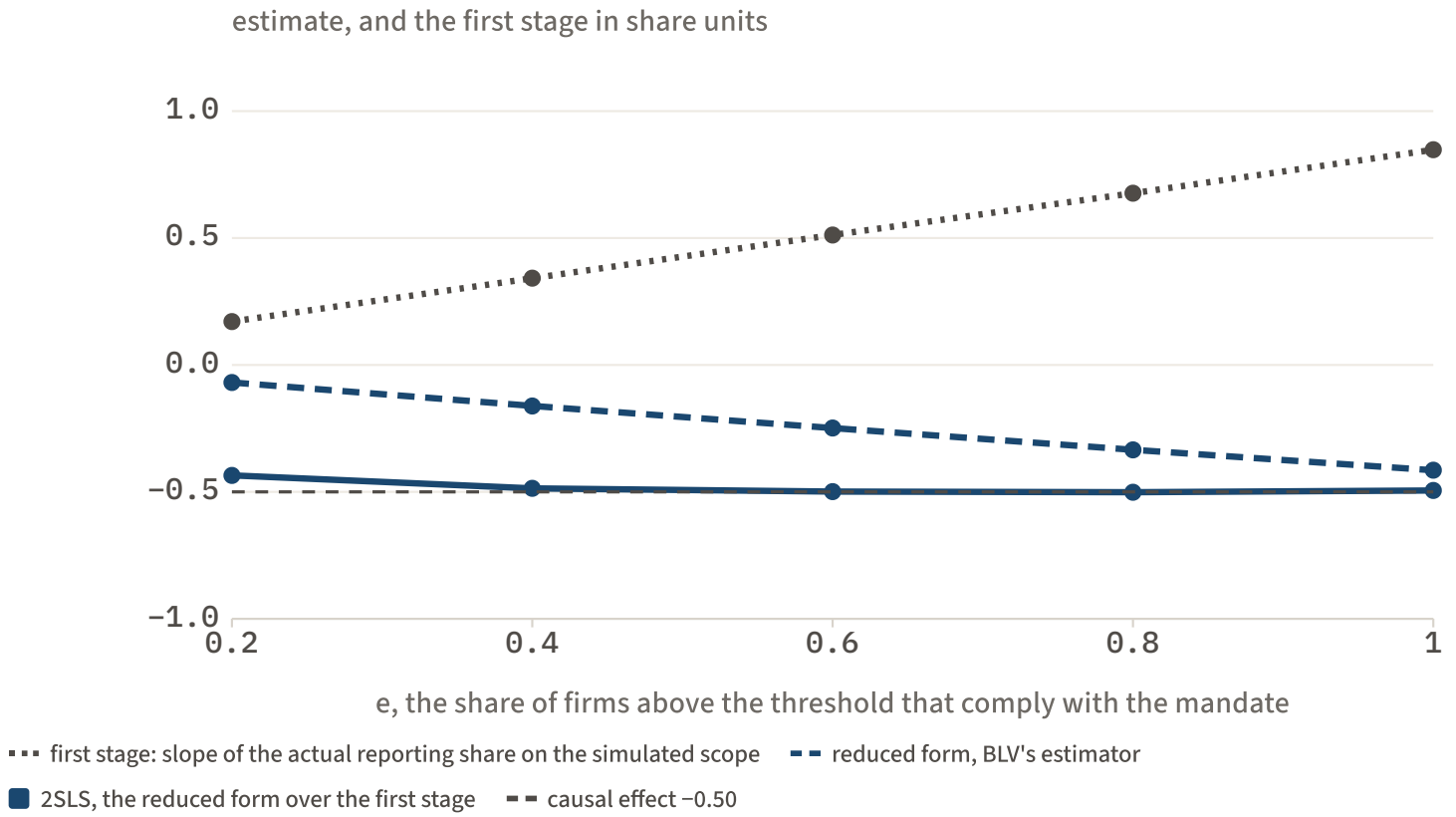
- **BLV regress innovation on the simulated scope and report no first stage, so β is the effect of a unit of statutory scope, the policy-relevant quantity; the effect per unit of actual reporting is β over the first stage, which is below one even at full compliance and falls with it.**

An exact identity, and its population counterpart with a constant effect

$$b_{2SLS} = \frac{b_{RF}}{\hat{\pi}}, \qquad \operatorname{plim} b_{RF} = \beta\,\pi$$

> *"the estimated reporting mandate effect in the EU setting reflects the average reporting format, exemption, and enforcement level across our sample countries, industries, and years."*
>
> Breuer, Leuz, and Vanhaverbeke (2025, fn. 9, p. 6)

At compliance 0.6 the first stage is 0.51, the reduced form −0.25, and 2SLS −0.50. The German reform (slide 18) is a compliance shock of this kind, from 5 to 10 percent to over 90 (p. 7).



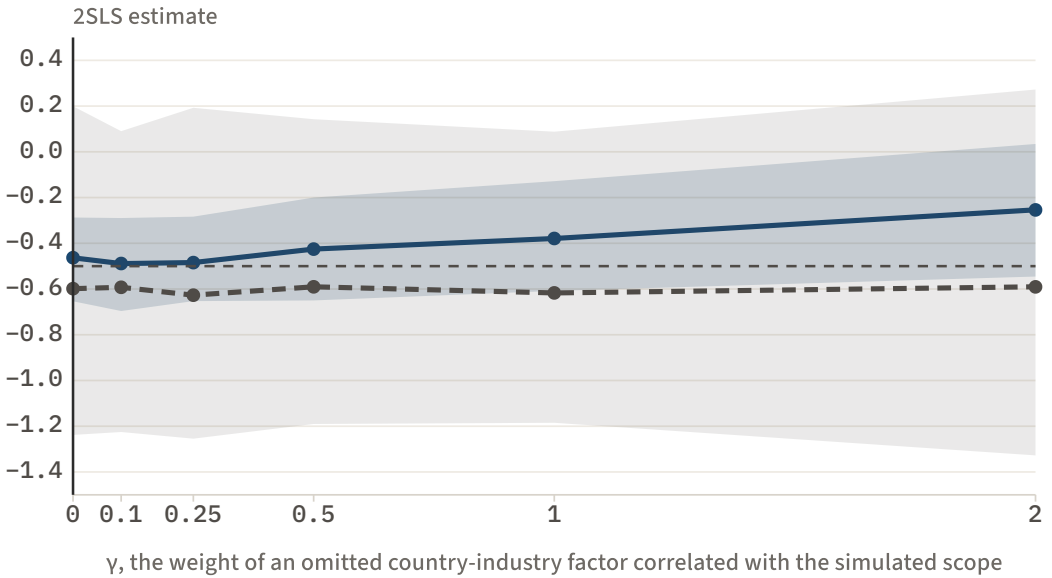
Simulated (seed 2025, 300 draws per point), reverse causality held at λ = 0.5: each firm above the threshold complies with probability e, the same in every country, so the actual reporting share is e times the share above the threshold and the reduced form is β times the first stage; with compliance that differs across countries, 2SLS would be a compliance-weighted average.

An omitted country-industry factor survives the paper's fixed effects

- **BLV's identifying assumption is that no omitted factor is correlated with both innovation and the intensity measure at the country-industry level. A bundled size-based rule that shares the thresholds, an audit exemption or a simplified format, is such a factor: at a weight of one it moves 2SLS to −0.38 and at two to −0.25, while the first stage stays at F ≈ 460.**

> *"Our identifying assumption is that there are no omitted factors correlated with corporate innovation and our intensity measure at the country-industry level."*
>
> Breuer, Leuz, and Vanhaverbeke (2025, p. 9)

Two routes to the claim (slide 04, shares or shifts): the shares are exogenous given the effects (Goldsmith-Pinkham, Sorkin, and Swift 2020), or the shifts are as good as random across many shocks (Borusyak, Hull, and Jaravel 2022). BLV argue the shifts, thresholds set uniformly across industries (p. 9); with at most 26 countries the claim also needs the shares.



γ = 0.00 · paper's effects: 2SLS **−0.46**, F ≈ 462 · plus country-industry effects: 2SLS **−0.60** (interquartile range −1.24 to 0.20), F ≈ 45

γ 0

2SLS, country-year and industry-year effects (the paper) · its interquartile range · 2SLS, the same plus country-industry effects · its interquartile range · causal effect −0.50

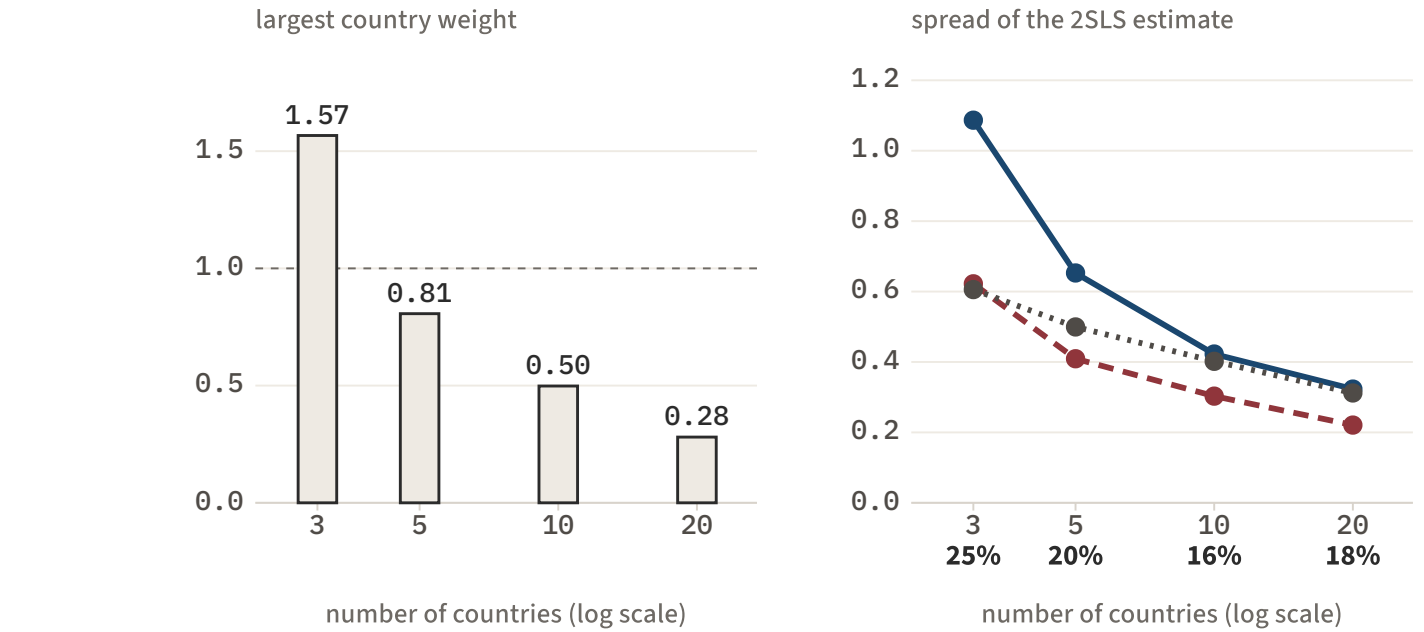
Simulated (seed 2025, 300 draws per point), reverse causality held at λ = 0.5: the outcome also contains γ times a time-invariant country-industry factor, the share of the industry's firms in the size bin just above the country's initial threshold (the incidence of a bundled size-based rule that shares the thresholds), which is not the mandate. The first stage is unaffected. Country-industry effects leave only the 3 countries whose threshold changes.

With few countries, the estimate rests on one or two threshold levels

- **The shifts route needs many as-good-as-random shocks. With 3 countries the largest country weight is 1.57 and the spread of 2SLS across draws 1.09; with 20 they are 0.28 and 0.32. The two-way clustered standard error BLV use understates the spread at every number of countries.**

> A country weight is this note's analogue of GPSS's Rotemberg weights: the coefficient on the just-identified estimate that uses only that country's variation; the weights sum to one and can be negative. BLV report none. Appendix: one draw's weights.

The simulated country factor loads on exposure, so residuals are correlated within a country across industries and years; clusters at country-industry and country-year do not contain that, clusters at the country, the level of the shock, do (Borusyak, Hull, and Jaravel 2022). BLV cluster two-way with 150 to 160 country-year clusters (Table 2, p. 12).



Simulated (seed 2025, 300 draws per point): a country-level factor, SD 0.3, whose effect on an industry scales with its share of small firms; the paper's effects; three countries change their threshold. Country weights sum to one and can be negative, so the largest can exceed one.

The paper meets some of the literature's diagnostics, not all

■ **Each row is a question the shift-share literature has learned to ask; the right column is what the published paper offers.**

Question	Source	In Breuer, Leuz, and Vanhaverbeke (2025)
Shifts as good as random, and many of them	Borusyak, Hull, and Jaravel (2022)	thresholds uniform across industries, motivated by fixed costs (p. 9); at most 26 countries, thresholds persistent (fn. 15)
Shares exogenous: pre-determined, balanced on confounders	Goldsmith-Pinkham, Sorkin, and Swift (2020)	shares pooled over 2000 to 2018, not a pre-period (fn. 16, p. 9); balance tests cited from Breuer (2021)
The shares and the countries that carry the estimate	Rotemberg weights, GPSS (2020)	not reported
The first stage behind the reduced form	Currie and Gruber (1996)	no first stage for Europe; the measure is the regressor (p. 9); fn. 9 gives the average-enforcement reading
Nonrandom simulated exposure, where a counterfactual assignment can be specified	Borusyak and Hull (2023), recentering	not addressed
Standard errors at the level of the shock or across similar units	BHJ (2022); Adão, Kolesár, and Morales (2019)	two-way clustering by country-industry and country-year
A second design with different assumptions	BLV pp. 9 to 10	the German 2007 enforcement reform (slide 18)

> *"If the EU or specific countries really intended to treat industries differently, they could have set at least some industry-specific exemption thresholds, but they chose not to do this."*
>
> Breuer, Leuz, and Vanhaverbeke (2025, p. 9)

The German enforcement reform supplies a second design

- **Germany did not enforce the mandate until the 2007 EHUG reform; BLV compare county-industries by their pre-period share of limited-liability firms before and after enforcement, a temporal difference-in-differences with continuous exposure whose assumption is parallel trends, not the exogeneity of thresholds.**

The German specification (p. 9): county-year, industry-year, and county-industry effects

$$Y_{d,i,t} = \beta \, \text{Limited}_{d,i} \times \text{post}_t + \alpha_{d,t} + \delta_{i,t} + \phi_{d,i} + \varepsilon_{d,i,t}$$

−0.132

on the share of innovating firms (t = −3.46); Table 6, p. 16. An interquartile-range rise in the limited-liability share (0.436 to 0.764, Table 1) is associated with about 4.3 points fewer innovating firms, 8 percent of the German mean of 0.551 (this note's arithmetic)

47,185

county-industry-years in Table 6, 8,190 county-industry clusters; more than 20,000 firms, 2002 to 2013 (p. 3)

Fig. 1 (p. 17, not reproduced here): event-time coefficients near zero in 2002 to 2006 and falling from 2008, which BLV read as support for parallel trends (p. 15). The reform is an enforcement shock, compliance from 5 to 10 percent to over 90, the real-world version of slide 14's compliance rate.

The credibility of a shift-share measure rests on its exogeneity claim

1. A shift-share measure keeps the part of a treatment that a fixed exposure and a policy produce. In the simulation the simulated scope omits the firms' size responses, so reverse causality reverses the sign of OLS and leaves the reduced form and 2SLS unmoved.
2. What the construction cannot remove is a factor correlated with the exposure-by-policy interaction. The claim has to be made for the shares or for the shifts, inside the fixed effects used; BLV argue the shifts, cite balance tests for the shares, and add a second design with a different assumption.
3. For BLV the burden falls on the shares, since 26 persistent thresholds are few shocks. For a size-threshold design of one's own, the questions are what plays the shares, what plays the shifts, and which of the two claims the institution supports.

Breuer, Leuz, and Vanhaverbeke (2025) · Goldsmith-Pinkham, Sorkin, and Swift (2020) · Borusyak, Hull, and Jaravel (2022) · github.com/sbb151/acctg597a-research-design

Appendix

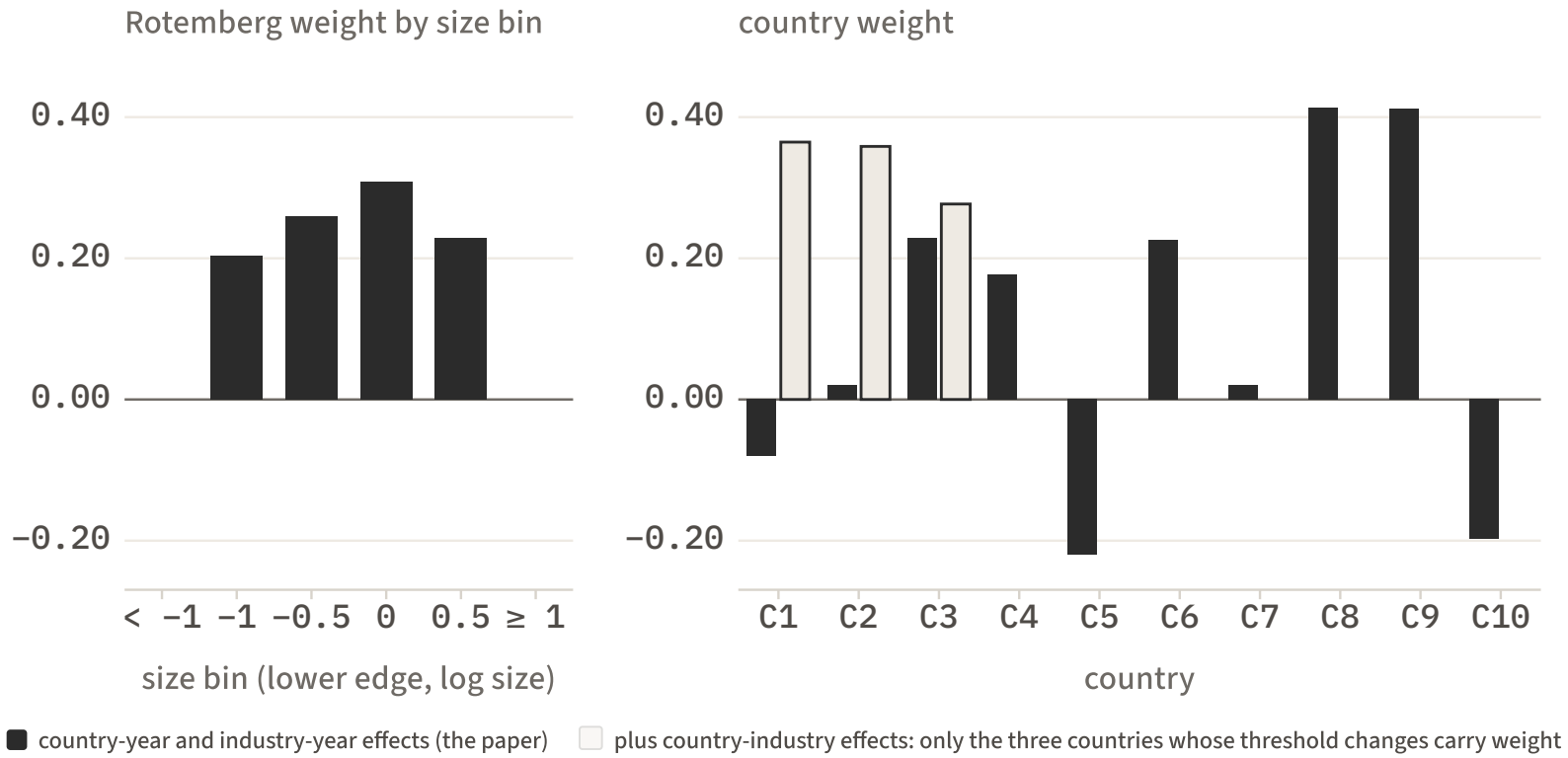


The algebra: the sum, the reduced form, and the decompositions

The measure. With size bins b and an industry's pooled shares s_{ib}, $\text{Reporting}_{cit} = \sum_b s_{ib}\, g_{bct}$ with $g_{bct} = 1[\text{bin } b \text{ above } \tau_{ct}]$. The simulation's bins are cut at the five threshold values, so the sum equals $1 - F_i(\tau_{ct})$ exactly.

The reduced form. With $\tilde{}$ for residuals from the fixed effects, $b_{RF} = \tilde{z}'\tilde{y}/\tilde{z}'\tilde{z}$, $\hat{\pi} = \tilde{z}'\tilde{x}/\tilde{z}'\tilde{z}$, and $b_{2SLS} = \tilde{z}'\tilde{y}/\tilde{z}'\tilde{x} = b_{RF}/\hat{\pi}$ (asserted to 10^{-8} against `ivregress` under both structures).

Rotemberg weights. Writing $\tilde{z} = \sum_b \tilde{z}_b$ with $\tilde{z}_b$ the demeaned $s_{ib}\, g_{bct}$, $b_{2SLS} = \sum_b \alpha_b\, b_b$ with $\alpha_b = \tilde{z}_b'\tilde{x}/\tilde{z}'\tilde{x}$ and $b_b = \tilde{z}_b'\tilde{y}/\tilde{z}_b'\tilde{x}$ (GPSS 2020); the α_b sum to one. Summing instead over countries, $\tilde{z}_c = \tilde{z} \cdot 1[\text{country } c]$, gives the country weights of slide 16.



One draw (seed 2025). Left: Goldsmith-Pinkham, Sorkin, and Swift's (2020) Rotemberg weights, one per share instrument (a size bin's share times its shift). Right: the same decomposition over countries, this note's analogue. Each weight multiplies the just-identified estimate that uses only that instrument; weights sum to one and can be negative; the weighted estimates sum to 2SLS (identity asserted in the script).

Back to the anatomy, which variation, the reduced form, or few countries.

The simulation's data-generating process

- **Base Stata, seed 2025 (`scripts/bartik_shift_share.do`, about six minutes); every number on the stages is a mean over 300 draws unless marked as one draw.**

- **Cells.** 30 industries, 10 countries, 8 years, 50 firms per cell; base log size $\ell_{i0} \sim N(\mu_k, 1)$ with μ_k evenly spaced on $[-1.2, 1.2]$; six size bins cut at $-1, -0.5, 0, 0.5, 1$.
- **Shifts.** Each country's threshold takes one of the five cut values, in random order so that levels differ; three countries move theirs by ± 0.5 once, in a random year between 3 and 6.
- **Shares.** s_{kb} is the share of the industry's base-period firms in bin b, pooled across countries. The simulated scope is $z_{kct} = \sum_b s_{kb}\, 1[\text{bin } b \geq \tau_{ct}]$.
- **Sizes and the actual share.** $\ell_{ict} = \ell_{i0} + \lambda u_{kct} + \nu_{ict}$ with $\nu \sim N(0, 0.3^2)$; a firm above the threshold complies with probability e; x_{kct} is the share of the cell's firms above the threshold that comply.
- **Outcome.** $y_{kct} = \alpha_k + \delta_{ct} + \beta x_{kct} + u_{kct} + \gamma\,(w_{kc} - \bar{w}) + \rho_c\,\text{small}_k + e_{kct}$ with $\beta = -0.50$, $u \sim N(0, 0.5^2)$, $e \sim N(0, 0.2^2)$; w_{kc} the share of industry k's firms in the bin just above country c's initial threshold, $\rho_c \sim N(0, \sigma_\rho^2)$ a country factor, small_k the share of base firms below log size 0.
- **Grids.** λ (slide 12); γ (slide 15), the number of countries with $\sigma_\rho = 0.3$ (slide 16), and e (slide 14), these three at λ = 0.5 with the other disturbances off.

Identity checks printed by the script: Stata's `regress` and `ivregress` agree with the demeaned ratios and with the Mata clustered standard error; 2SLS equals the reduced form over the first stage; the weights sum to one and the weighted estimates sum to 2SLS; with country-industry effects added, the level part of z demeans to zero. Back to exposure,

Pooled shares, managed sizes, and a correction

- **GPSS's shares argument supposes pre-determined exposure; BLV's distributions pool every sample firm-year, so the shares are fixed across countries and years once built, which is not the same as pre-determined.**

> *"We also include all firm-year observations in the construction of the standardized size distribution, not just firm-years after 2007."*
>
> Breuer, Leuz, and Vanhaverbeke (2025, fn. 16, p. 9)

- **Sizes shaped by the mandate.** The Europe design has no pre-period; every firm-year in the pooled distribution was observed under some country's thresholds, so the shares are an equilibrium object. BLV's argument is that the pooled distribution "does not vary across countries … or over time" (p. 9); that pooling across 26 countries and 19 years dilutes any one country-year's response is this note's reading.
- **Managed sizes.** Bernard, Burgstahler, and Kaya (2018) document size management below EU thresholds; a pooled distribution then carries a notch just below the common thresholds, so the shares near the thresholds are themselves a response to the policy.
- **A computed object.** The shares cannot be inspected without their construction; fn. 16 follows Breuer (2021) "and corrected in Breuer (2025)," a published correction of program-code errors in the scope variables (JAR 63(2): 1031–1055).

None of the three is a demonstrated bias; the simulation keeps its shares pre-period and clean, so its 2SLS is unbiased by construction. They are reasons a first stage and the weights are worth asking for. Back to the diagnostics.